Terraform Task-2

Task Description:

Create 2 EC2 instances on 2 different regions and install nginx using terraform script.

Installed AWS CLI and Terraform:

```
GHOST@GHOST: ~/Projects/[
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$ aws --version

aws-cli/2.34.2 Python/3.13.11 Linux/6.6.87.2-microsoft-standard-WSL2 exe/x86_64.ubuntu.24
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$ terraform --version
Terraform v1.14.8
on linux_amd64
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$ ls
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$
```

Creating a terraform script file to launch 2 EC2 instances on different regions and install nginx using the same file:

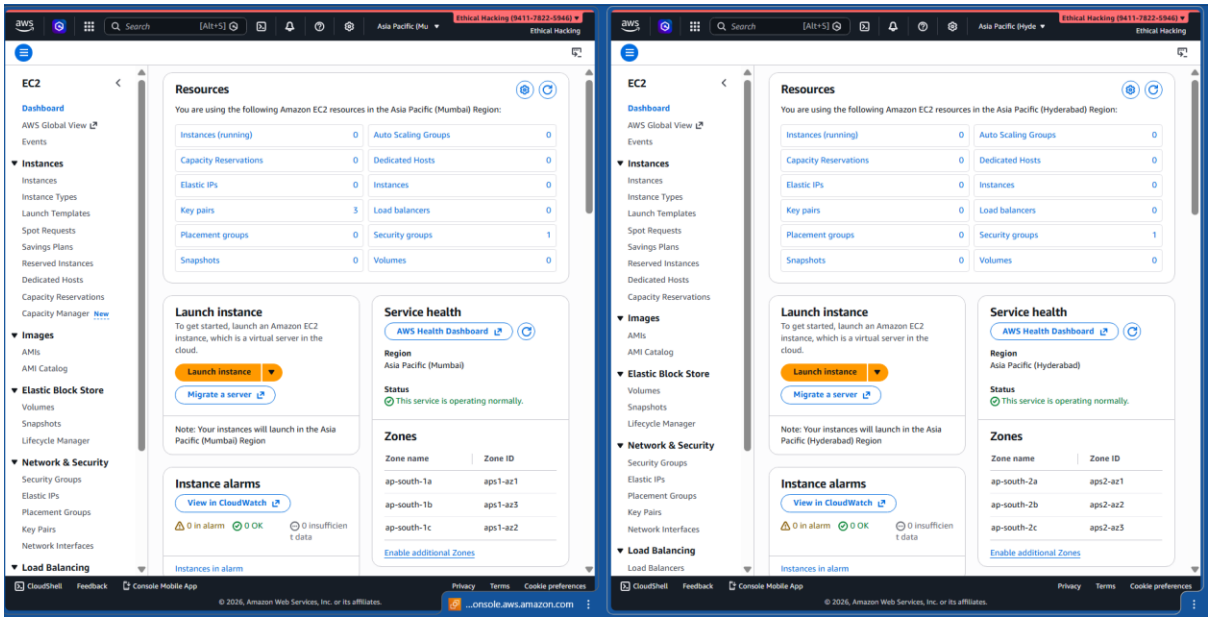
```
GHOST@GHOST: ~/Projects/[
# PROVIDERS (Multi-region)

provider "aws" {
  alias  = "mumbai"
  region = "ap-south-1"
}

provider "aws" {
  alias  = "hyderabad"
  region = "ap-south-2"
}

# FETCH LATEST UBUNTU AMI - MUMBAI

data "aws_ami" "ubuntu_mumbai" {
  provider    = aws.mumbai
  most_recent = true

  owners = ["099720109477"] # Canonical

  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }
}

# FETCH LATEST UBUNTU AMI - HYDERABAD

data "aws_ami" "ubuntu_hyderabad" {
  provider    = aws.hyderabad
  most_recent = true

  owners = ["099720109477"]

  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }
}

# SECURITY GROUP - MUMBAI

resource "aws_security_group" "sg_mumbai" {
  provider = aws.mumbai

  name        = "allow_http_mumbai"
  description = "Allow HTTP and SSH"

  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
"main.tf" 155L, 2868B
```

```
GHOST@GHOST: ~/Projects/[
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}

# SECURITY GROUP - HYDERABAD

resource "aws_security_group" "sg_hyderabad" {
  provider = aws.hyderabad

  name        = "allow_http_hyderabad"
  description = "Allow HTTP and SSH"

  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}

# EC2 INSTANCE - MUMBAI

resource "aws_instance" "mumbai_ec2" {
  provider = aws.mumbai
```

GHOST@GHOST: ~/Projects/[

```
# EC2 INSTANCE - MUMBAI

resource "aws_instance" "mumbai_ec2" {
  provider = aws.mumbai

  ami           = data.aws_ami.ubuntu_mumbai.id
  instance_type = "t3.micro"

  vpc_security_group_ids = [aws_security_group.sg_mumbai.id]

  user_data = <<-EOF
              #!/bin/bash
              apt update -y
              apt install nginx -y
              systemctl start nginx
              systemctl enable nginx
              EOF

  tags = {
    Name = "Nginx-Mumbai"
  }
}

# EC2 INSTANCE - HYDERABAD

resource "aws_instance" "hyderabad_ec2" {
  provider = aws.hyderabad

  ami           = data.aws_ami.ubuntu_hyderabad.id
  instance_type = "t3.micro"

  vpc_security_group_ids = [aws_security_group.sg_hyderabad.id]

  user_data = <<-EOF
              #!/bin/bash
              apt update -y
              apt install nginx -y
              systemctl start nginx
              systemctl enable nginx
              EOF

  tags = {
    Name = "Nginx-Hyderabad"
  }
}

# OUTPUTS

output "mumbai_public_ip" {
  value = aws_instance.mumbai_ec2.public_ip
}
```

Before running the script:



Initiating, planning and applying the script:

```
GHOST@GHOST: ~/Projects/[

GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.38.0...
- Installed hashicorp/aws v6.38.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$
```

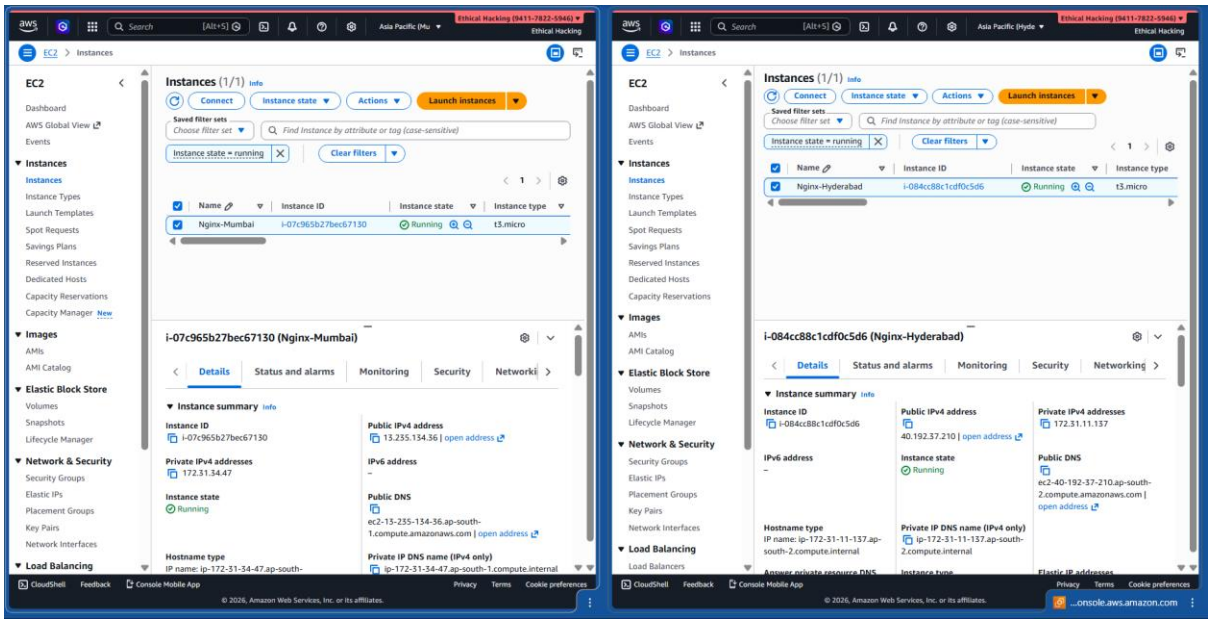
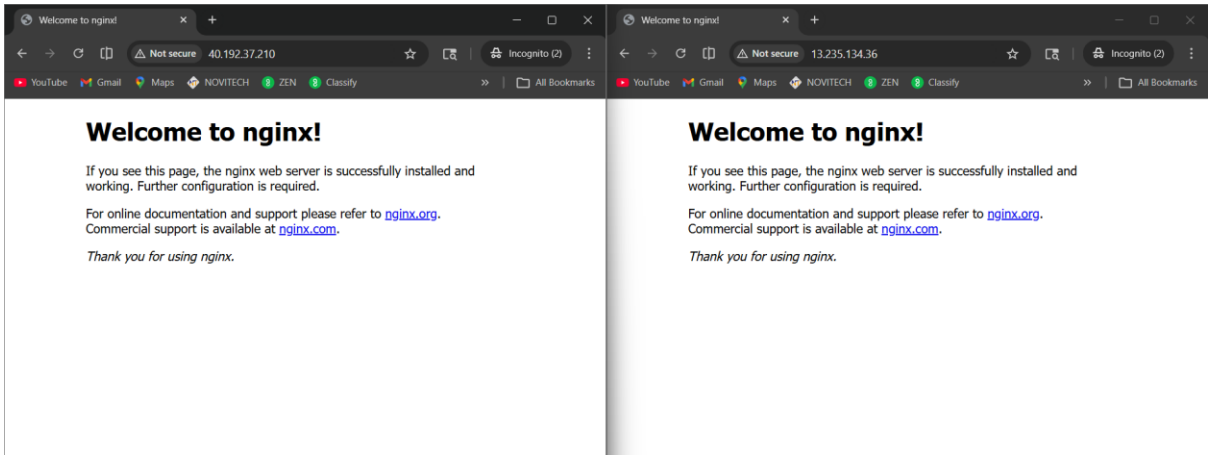
```
commands will detect it and remind you to do so if necessary.
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$ terraform plan
data.aws_ami.ubuntu_mumbai: Reading...
data.aws_ami.ubuntu_hyderabad: Reading...
data.aws_ami.ubuntu_hyderabad: Read complete after 1s [id=ami-04d1c5d5066b58e37]
data.aws_ami.ubuntu_mumbai: Read complete after 1s [id=ami-095cc90aa5ddff518]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.hyderabad_ec2 will be created
  + resource "aws_instance" "hyderabad_ec2" {
      + ami                                  = "ami-04d1c5d5066b58e37"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "ap-south-2"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
```

```
GHOST@GHOST: ~/Projects/I

GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$ terraform apply -auto-approve
data.aws_ami.ubuntu_hyderabad: Reading...
data.aws_ami.ubuntu_mumbai: Reading...
data.aws_ami.ubuntu_hyderabad: Read complete after 1s [id=ami-04d1c5d5066b58e37]
data.aws_ami.ubuntu_mumbai: Read complete after 1s [id=ami-095cc90aa5ddff518]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.hyderabad_ec2 will be created
  + resource "aws_instance" "hyderabad_ec2" {
      + ami                                  = "ami-04d1c5d5066b58e37"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "ap-south-2"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "Nginx-Hyderabad"
```

```
GHOST@GHOST: ~/Projec

            + prefix_list_ids  = []
            + protocol         = "tcp"
            + security_groups  = []
            + self             = false
            + to_port          = 22
              # (1 unchanged attribute hidden)
          },
        + {
            + cidr_blocks      = [
                + "0.0.0.0/0",
              ]
            + from_port        = 80
            + ipv6_cidr_blocks = []
            + prefix_list_ids  = []
            + protocol         = "tcp"
            + security_groups  = []
            + self             = false
            + to_port          = 80
              # (1 unchanged attribute hidden)
          },
      ]
    + name                   = "allow_http_mumbai"
    + name_prefix            = (known after apply)
    + owner_id               = (known after apply)
    + region                 = "ap-south-1"
    + revoke_rules_on_delete = false
    + tags_all               = (known after apply)
    + vpc_id                 = (known after apply)
  }

Plan: 4 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + hyderabad_public_ip = (known after apply)
  + mumbai_public_ip    = (known after apply)
aws_security_group.sg_mumbai: Creating...
aws_security_group.sg_hyderabad: Creating...
aws_security_group.sg_hyderabad: Creation complete after 3s [id=sg-0af5102e27258de36]
aws_security_group.sg_mumbai: Creation complete after 3s [id=sg-0e565e0acd57535da]
aws_instance.hyderabad_ec2: Creating...
aws_instance.mumbai_ec2: Creating...
aws_instance.hyderabad_ec2: Still creating... [00m10s elapsed]
aws_instance.mumbai_ec2: Still creating... [00m10s elapsed]
aws_instance.hyderabad_ec2: Creation complete after 13s [id=i-084cc88c1cdf0c5d6]
aws_instance.mumbai_ec2: Creation complete after 13s [id=i-07c965b27bec67130]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

Outputs:

hyderabad_public_ip = "40.192.37.210"
mumbai_public_ip = "13.235.134.36"
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$
```

```
GHOST@GHOST: ~/Projects/[

GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$ ls -la
total 48
drwxr-xr-x  4 GHOST GHOST  4096 Mar 26 11:21 .
drwxr-xr-x 21 GHOST GHOST  4096 Mar 26 10:22 ..
drwxr-xr-x  3 GHOST GHOST  4096 Mar 26 11:14 .terraform
-rw-r--r--  1 GHOST GHOST  1377 Mar 26 11:14 .terraform.lock.hcl
drwxr-xr-x  2 GHOST GHOST  4096 Mar 26 11:22 OutputSnips
-rw-r--r--  1 GHOST GHOST  2868 Mar 26 11:11 main.tf
-rw-r--r--  1 GHOST GHOST 23206 Mar 26 11:16 terraform.tfstate
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_18$
```

Verifying the creation of EC2 instances in both regions:



Verifying the installation of Nginx in the Console